

Noah Hart

Gameplay Programmer | Designer

CONTACT

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SUMMARY

Gameplay Programmer specializing in Unreal Engine 5 systems, AI behaviors, and modular gameplay architecture. Experienced leading engineering sprint planning and implementing combat and encounter systems in team-based development environments. Strong foundation in C++ design patterns and rapid prototyping across multiple engines.

TECHNICAL SKILLS

- **Programming Languages**
 - C, C++, C#, Python, Lua, JavaScript, HTML, Unreal Blueprints
- **Engines/Tools**
 - Unreal Engine 5 (C++ & Blueprints), Unity, Phaser3, Blender, Jira, Trello, Git(hub), Perforce, Adobe Suite, openframeworks, Visual Studio
- **Gameplay and Systems**
 - Gameplay Systems
 - Combat Design & Enemy/Boss Design
 - AI Behavior Logic (Behavior/Decision Trees, GOAP)
 - State Machines - Animations
 - Design Patterns (State, Observer, Command)
 - Sprint-based Production
 - Rapid Prototyping
- **General**
 - Teamwork & Collaboration
 - Leadership & Active Listening
 - Problem Solving & Critical Thinking

EDUCATION

University of California Santa Cruz

Bachelors of Computer Science - 3.8 GPA

2019 - 2023

Coursework: Data Structures, Computer Graphics, AI, Game Design & Development

University of Utah

Masters of Entertainment Arts and Engineering - Engineering Track

2024 - 2026

Worked in a simulated studio environment using sprint-based production, cross-disciplinary collaboration, and milestone-driven development workflows.

IMPORTANT LINKS

- **Portfolio:** <https://nrhart.github.io/NoahsPortfolio/>
- **Github:** <https://github.com/nrhart>
- **Itch:** <https://nrhart.itch.io/>
- **Game Steam Page:** <https://store.steampowered.com/app/4125300/NoRail/>

EXPERIENCE & PROJECTS

Lead Engineer, Systems & Level Designer

NO RAIL

Archie Games - GAMES Studio | 2025-2026

- Architected modular gameplay systems in Unreal Engine 5
- Led engineering sprint planning and system integration
- Designed and implemented level layout and core encounter logic
- Collaborated cross-discipline to deliver a thesis-scale build in a studio simulated environment

GRAPHICS ENGINE DEVELOPMENT

University of Utah | 2025

- Modified an existing graphics engine to support platform-independent rendering through abstraction interfaces that decoupled platform-specific graphics APIs
- Developed a real-time game within the engine to validate rendering, input, physics, UI, and audio system integration

AI SYSTEMS & PATHFINDING PROJECTS (OPENFRAMEWORKS)

University of Utah | 2026

- Implemented A* and Dijkstra pathfinding with multiple heuristics (Manhattan, Euclidean, randomized) and analyzed performance tradeoffs
- Built grid-based navigation system and integrated real-time path following agents
- Developed procedural maze generation (PCG) system used to create a Pac-Man-style game environment
- Evaluated algorithm efficiency using node visitation metrics and heuristic behavior

C++ GAMEPLAY SYSTEMS & COMBAT DESIGN - UNREAL ENGINE 5

University of Utah | 2025-2026

- Designed and implemented modular gameplay and combat systems in Unreal Engine 5 using C++ and object-oriented architecture.
- Applied industry-standard design patterns (State, Observer, Command) to gameplay systems and player state logic.
- Developed event-driven gameplay frameworks supporting scalable combat encounters and gameplay interactions.
- Built CineCaster, a UE5 plugin for real-time cinematic camera control in multiplayer with dynamic player switching and smooth camera blending.
- Collaborated on gameplay prototyping and combat-focused systems emphasizing responsiveness, extensibility, and iteration speed.

TEACHING ASSISTANT

University of Utah | 2025-2026

- Led a team of TAs supporting 100+ students in Unreal Engine C++ development
- Provided debugging assistance on gameplay systems and architecture
- Mentored capstone teams in delivering vertical slice builds

RAPID PROTOTYPING & ALT CONTROL

University of Utah | 2024-2025

- Designed and implemented three gameplay prototypes within 3-4 week development cycles across Unity and Unreal
- Served as hardware engineer, developing and integrating custom physical input systems with Unity gameplay logic
- Built unconventional control mechanics, including transforming a rocking horse into a functional kart racing controller
- Nominated at Indiecade and showcased at University Launch Event

RAPID PROTOTYPING & CAPSTONE

University of California Santa Cruz | 2022-2023

- Used physical and digital prototyping methods to rapidly develop 3 game projects. Developed in Unity, Unreal, and Phaser (HTML+JavaScript).
- Served in Team Lead and Producer roles, practicing scheduling and coordination methods.